SECTION 13009-A
CLEANROOM CERTIFICATION

PART 1 – GENERAL

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1.2 WORK INCLUDED

A. This Section specifies services required of a cleanroom testing and certification agency, hereinafter referenced as the cleanroom certification contractor (CCC), to measure and record the conditions prevailing at the completion of cleanroom construction fit-up in the as-built condition or various stages in which testing finally commences.

B. The CCC shall retest all repairs, replacements, and adjustments dictated by failure of materials or performance to meet acceptance criteria.

C. Work described in the table following under "benchmark tests" shall be bid with separate prices for each test to permit the Owner to select specific work which will create baseline facility performance evaluation at startup.

D. The CCC shall assist the testing, adjusting, and balancing contractor (TABC) to achieve a final cleanroom balance. This shall include advising on the adjustment and setting of raised floor panels and floor and filter dampers.

E. The CCC shall act on the Owner’s behalf in any technical discussion or negotiation involving deficiencies, repairs, and/or corrections. Such actions shall be initiated only by the director.

F. If, during the cleanroom certification, a discrepancy between reported balance values, as reported by the TABC and as measured by the CCC arise, the CCC will be considered an authorized agent of the Owner. The TABC shall correct any deficiencies observed by the CCC. The final authorities for resolution of disputes concerning balance issues shall be the director and the Construction Manager (CM).

G. Test procedures have been structured to measure and document that cleanroom areas and support systems perform in accordance with project design criteria.

1.3 RELATED WORK

A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for the certification of the cleanrooms:
   1. Division 1 sections included in the project specifications.
   2. The Contract.
   3. Section 01335 - Clean Zone General Requirements.
   4. Section 13004 - Cleanroom Zone Systems.
   5. Section 13007 - Clean Zone Cleaning Procedures
   6. Section 15050 - Materials and Methods.
   7. Section 15950 - Air and Water Systems Testing, Adjusting, and Balancing.

B. In the event of conflict regarding the requirements for certification listed between this Section and any other section, the provisions of this Section shall govern.

C. Air systems serving the cleanrooms will be initially tested, adjusted, and balanced under the TAB contract. The CCC shall work with the TAB to attain the final balance.

D. Repairs required to meet the acceptance criteria of this document shall be completed by the installing contractor.

1.4 SCOPE OF CERTIFICATION TESTS

A. For purposes of conformity, the same paragraph subsection identification letters have been used for the test instruments specified under paragraph 2.3 as for their respective test procedures specified under paragraph 3.3.

B. The CCC shall perform specific acceptance and benchmark tests in the rated cleanrooms described in Table T-1 below. Refer to specification paragraph 2.3, Instrumentation, and paragraph 3.3, Certification Procedures and Field Tests, where the subsection paragraphs are described.

Table T-1
Cleanroom Certification Tests

Subsection	Tests	ISO Class 3	ISO Class 5	ISO Class 6 and ISO Class 7
Final Acceptance Tests				
A	Bench filter leakage	By manufacturer	By manufacturer	NR
B	Ceiling filter leakage scan	1	1	NR
C	Airborne particle counts	1	1	1
D	Baseline airborne particle counts	NR	NR	NR
E	Airflow volume and velocity	1	1	NR
F	Airflow parallelism	1	NR	NR
G	Room pressurization	1	1	1
H	Enclosure induction leak	NR	NR	NR
I	Room recovery rate	NR	NR	NR
J	Makeup air handler HEPA/ULPA final filter media	1	1	NR
K	Environmental uniformity	1	1	1
L	Noise level	1	1	1
M	Floor vibration	NR	NR	NR
N	Lighting level	1	1	1
O	Light wave spectrum (yellow light areas)	1	1	1
P	Floor conductivity	1A	1	NR

Test Requirement Code:
NR = Test is not required by this Section.
1 = Test is required in the as-built facility phase.
1A = Test is desired in the as-built facility phase, but indicate separate alternative cost for each of the acceptance tests shown.

1B = Test is desired in the as-built facility phase, but indicate separate alternative cost for each of the benchmark tests shown.

C. The CCC shall propose per diem rates for labor and expenses to apply to additional field work required beyond the contracted scope of work.

D. At completion of this as-built certification work, the Owner may engage the services of a testing and certifying agency, hereinafter referenced as the hookup cleanroom certification contractor (HCCC), to measure and record the cleanroom conditions affected by ongoing process tool move-in, installation, and hookup. That work will continue through the at-rest and operating cleanroom occupancy phases as required.

1.5 REFERENCES

A. Federal Standard 209, Latest Edition: Airborne Particulate Cleanliness Classes in Cleanrooms and Clean Zones.

General Services Administration
Specifications Activity
Printed Materials Supply
Building 197, Naval Weapons Plant
Washington, DC 20407
(708) 255-1561

B. IES-RP-CC-006 - Testing Cleanrooms, 1995:
1. IES-RP-CC-007: Testing ULPA Filters (1992).
2. IES-RP-CC-024: Measuring and Reporting Vibration (1993).

Institute of Environmental Sciences
940 East Northwest Highway
Mount Prospect, IL 60056

C. Current edition of practices, methods, or standards as prepared by the following technical societies and associations. Where conflicts exist between standards, the Contractor shall conform to the more severe requirements as defined by the director.
1. AABC: National Standard for Field Measurements and Instrumentation, Total System Balance, Volume 1.
2. AMCA Publication 203: AMCA Fan Applications Manual - Field Performance Measurements.
3. ANSI S1.13-1976: Methods for Measurement of Sound Pressure Levels.
4. ANSI S12.2: Procedure of Measuring and Rating Steady-State Room Noise.
5. ASHRAE: 1992 Systems Handbook.

6. CSI: Construction Specification Institute Standards.
7. ASHRAE 41.3-89: Standard Method for Pressure Measurement.
8. ASHRAE 41.4-86: Standard Method for Temperature Measurement.
9. ASHRAE 111-88: Practices for Measurement, Testing, Adjusting, and Balancing of Building Heating, Ventilation, Air Conditioning, and Refrigeration Systems.
10. ASME PTC 19.2-64, Part 2: Pressure Measurement Instruments and Apparatus.
11. ASME PTC 19.2-74, Part 3: Temperature Measurement Instruments and Apparatus.
12. ASTM 50-69: Continuous Sizing and Counting of Airborne Particles in Dust-Controlled Areas Using Instruments Based on Light-Scattering Principals.
13. ASTM F25-83: Sizing and Counting Airborne Particulate Contamination in Clean Rooms and Other Dust-Controlled Areas Designed for Electronic and Similar Applications.
14. NEBB: Procedural Standards for Certified Testing of Cleanrooms.
15. SMACNA: 1993, HVAC Systems - Testing, Adjusting, and Balancing.
16. SMACNA: Manual for the Balancing and Adjustment of Air Distribution Systems.

D. Definitions:
1. Refer to Section 01355, Clean Zone General Requirements, for a complete list of standard industry definitions applicable to this project.
2. As-Built Facility: cleanroom which is complete and operating with all services connected and functioning but has no production equipment or operating personnel in the room.
3. At-Rest Facility: cleanroom which is complete and has production equipment installed and operating but has no personnel within the room.
4. Balancing Agency: the Contractor's air and piping systems testing and balancing contractor (TABC) retained by the CM to assist and verify completion and performance of all operating systems.
5. Certification Contractor: the cleanroom testing and certification agency or contractor (CCC) retained by the Owner to verify final completion and performance of all cleanroom systems in the as-built mode.
6. Director: the person vested with authority by the Owner and the CM to supervise all construction personnel working within the Clean Zone to ensure requirements of the Contract Documents are met in completion of the Clean Zone construction, including all spaces, materials, and utilities within. The director will be retained and managed by the CM

performing the services specified in Section 13004, Cleanroom Zone Systems.

7. Hookup Certification Contractor: the testing and certification agency or contractor (HCCC) retained by the Owner to validate completed process tool hookup in the at-rest and operating modes.
8. Operating Cleanroom: cleanroom in normal operation with all services functioning and with production equipment and personnel present and performing their normal work functions within the room.

1.6 DESIGN CRITERIA

A. The following requirements have been extracted from the project design criteria. These criteria provide guidance to the CCC regarding the design intent of the facility support systems to create proper environmental conditions.

1. Class M3.5 (100) Fab Area:
    a. Operating Particle Limit: refer to Drawings.
    b. As-Built Particle Limit: refer to Drawings.
    c. Average Room Air Velocity: refer to Drawings.
    d. ULPA Filter Efficiency: refer to Drawings.
    e. Vibration Control: refer to Section 09090.xls and Drawings.
    f. Temperature: refer to Section 09090.xls and Drawings.
    g. Relative Humidity: refer to Section 09090.xls and Drawings.
    h. Lighting: refer to Section 09090.xls and Drawings.
2. Class M4.5 (1,000) Support Areas:
    a. Operating Particle Limit: refer to Drawings.
    b. At-Rest Particle Limit: refer to Drawings.
    c. Average Room Air Velocity: refer to Drawings.
    d. HEPA Filter Efficiency: refer to Drawings.
    e. Temperature: refer to Section 09090.xls and Drawings.
    f. Relative Humidity: refer to Section 09090.xls and Drawings.
    g. Lighting: refer to Section 09090.xls and Drawings.

B. The following criteria have been extracted from the latest printing of Federal Standard 209; this data has been used in the development of acceptance criteria for this project as shown in Table T-3.

Table T-3
Airborne Particulate Cleanliness Classes

Class limits are given for each class name. The limits designate specific concentrations (particles per unit volume) of airborne particles with sizes equal to and larger than the particle sizes shown.

Class		Class Limits									
		Size 0.02μ Volume		Size 0.10 μ Volume		Size 0.20μ Volume		Size 0.30μ Volume		Size 0.50μ Volume	
SI	English	Cubic Meters	Cubic Feet	Cubic Meters	Cubic Feet	Cubic Meters	Cubic Feet	Cubic Meters	Cubic Feet	Cubic Meters	Cubic Feet
M0	--	175	5	35	0.99	--	--	--	--	1.0	0.03
M1		7060	200	350	9.91	75.7	2.14	30.9	0.88	10.0	0.28
M1.5	(1)	--	--	1,240	35.00	265	7.50	106	3.00	35.3	1.00
M2		70,600	2,000	3,500	99.10	757	21.40	309	8.75	100.0	2.83
M2.5	(10)			12,400	350	2,650	75	1,060	30.00	353	10
M3				35,000	991	7,570	214	3,090	87.50	1,000	28.3
M3.5	(100)			--	--	26,500	750	10,600	300	3,530	100
M4				--	--	75,700	2,140	30,900	875	10,000	283
M4.5	(1,000)			--	--	--	--	106,000	3,000	35,300	1,000
M5				--	--	--	--	309,000	8,750	100,000	2,830
M5.5	(10,000)			--	--	--	--	--	--	353,000	10,000
M6.5	(100,000)			--	--	--	--	--	--	3,530,000	100,000
Gray	N/A	Unclassified area served by residual clean supply and return air patterns									

1.7 QUALITY ASSURANCE

A. Cleanroom air systems shall be tested and certified by a qualified firm specializing in cleanroom certification. The CCC and key staff shall be in current and active status in good standing with the Institute of Environmental Sciences (IES) and shall be National Environmental Balancing Bureau (NEBB) certified. The CCC shall work closely with the construction trades as required to complete construction of the cleanrooms in accordance with the Contract Documents.

B. The CCC shall assume responsibility for field correlation of all measurement instruments to be used in the cleanroom. He shall notify the Owner, the director, the Engineer, the ceiling and filter vendors, the control contractor, the TABC, and any other organization which might perform subsequent validation or performance tests using its own instruments after completion of the certification service. At the designated time and place, instruments shall be reconciled or calibrated as recommended by the CCC and approved by the majority of the firms participating. Readings taken with equipment not qualified by this process will be considered invalid and unsuitable for test results or verification.

C. Measurement and test equipment used for field readings shall be the most current generation. Sampling shall be based upon accepted industry sampling and statistical procedures.

D. Equipment shall be calibrated in accordance with manufacturer's requirements at the beginning of certification work and be traceable by serial number to the National Institute for Standards and Technology (NIST) within the previous 6 months. Any test equipment with an upcoming recalibration due date that falls within this project work schedule shall be recalibrated before any tests are performed with the instrument.

E. The terms "or as approved" and "or equivalent" will be considered only if requested by the CCC in writing to the Owner with the proposal.

F. The reference standards for field tests and project record documents shall be the IES RP-CC-006, Testing Cleanrooms, and NEBB Procedural Standards for Certified Testing of Cleanrooms.

G. In case of conflict between the IES, NEBB, or these Specifications, this project specification shall prevail. In event of conflict between the NEBB Procedural Standards and IES Recommended Practice, conflicts shall be noted by the CCC and submitted for resolution to the director before commencing with the work.

1.8 QUALIFICATIONS OF CLEANROOM CERTIFICATION CONTRACTOR

A. Any firm proposing on this service shall have been in business a minimum of 5 years, specializing in cleanroom testing and certification work, and it must have successfully completed a minimum of five similar projects during this time. A list shall be provided showing projects similar in size, complexity, and cleanliness classification to this project that the firm has completed. This list shall include the project name, description of mechanical system, range of services provided, and the name and phone number of the design consultant and the Owner who were responsible for final acceptance of the certifying service and the CCC's project manager and field engineers.

B. The Project Manager for the CCC shall hold a management position within the firm and have a minimum of 5 years of combined engineering education and experience testing and certifying cleanrooms as a field engineer, as well as recent experience over 2 consecutive years and at least two previous projects acting as the project manager. The Project Manager shall supervise all aspects of field measurement, select appropriate test methods, consult with the Owner and the Engineer on technical matters, coordinate procedures with the director, and approve the final test report. The Project Manager shall have a thorough knowledge of theoretical and practical aspects of cleanroom design and operation and shall demonstrate procedures to correct cleanroom performance deficiencies. A list shall be available upon request of projects similar to this project that the Project Manager has managed and/or certified.

C. The field engineer shall have a minimum of 2 years of experience testing and certifying cleanrooms as a field engineer or field technician. The field engineer shall supervise field technicians assigned to complete the testing and certifying of the work, shall have a basic knowledge of cleanroom systems and adjustments, and shall be responsible for onsite testing and data acquisition. No field tests shall be taken without the field engineer's jobsite presence.

D. Field technicians shall have completed previous training in cleanroom operations and certifying procedures, shall have a thorough, demonstrable knowledge of test procedures and equipment, shall have worked in this capacity on at least one other similar project, and shall only perform field work under direct supervision of the field engineer.

1.9 CERTIFICATION CONTRACTORS

A. The following certification firms have already demonstrated proficiency to the satisfaction of the Owner to execute certification services on similar projects of this type, and they will be invited to tender a proposal:
   1. Biocon Inc., North Carolina, (919) 781-9777.
   2. Cleanroom Sciences, Arizona, (602) 278-7318.
   3. Dryden Engineering, California, (408) 727-7321.

B. The proposal for certification shall be tendered in two separately sealed envelopes and be divided into:

1. Description of technical requirements, method statements, references, and resumes required by this Specification.
2. Priced service proposal with all commercial issues, alternative bids, and exceptions to the scope of work described.

C. The Owner retains the right to approve final contractor selection and to select appropriate performance test alternative bids tendered as part of the proposal.

1.10 COORDINATION

A. The field engineer for the CCC shall visit the jobsite at least four times during strategic construction phases for the period that the finished cleanroom envelope is being constructed to become familiar with utility systems, air management scheme, system layouts, and finished materials.

B. Site visits, instrument field correlation, and certification activities shall be confirmed in writing with the director.

C. The field engineer shall coordinate activities with the work of other contractors who perform air handler adjustments, air distribution system balancing, electrical system activation, and similar work activities to ensure utility support systems comply with the acceptance criteria described for the cleanrooms.

D. The Project Manager shall schedule work activities closely with the Owner and the director so that crucial tests get completed in a timely sequence which will permit selective partial occupancy for tool move-in as may be essential for the Owner’s overall project schedule.

1.11 SUBMITTALS

A. Provide the following with the Bid:
   1. Qualifications of the CCC:
      a. Include a short chronicle of the history, staff size, ownership, and milestones in the firm's development.
      b. A list of project names, description of cleanrooms, range of certification services, and the name and phone number of the design consultant or owner responsible for final inspection and acceptance of the certification service.
   2. Qualifications of field technicians, the field engineer, and the Project Manager proposed for this project.

   a. Include name, assignment, expected duties, work experience, and advanced training.
   b. Provide a list of projects managed by the Project Manager similar in size and scope to this project.
   c. Provide documentation that the CCC's key staff are in current active status with IES and NEBB.
3. Written method statement outlining the testing and certification procedures and the specific sequence to be performed for the final acceptance and benchmark performance tests on this project. Indicate test procedures planned for particle counting, whether statistical sample or sequential sample counting.
4. A sample test report of a similar project for review by the Engineer only to verify the CCC's expertise collecting, interpreting, and recording data.
5. Description of instrumentation and test equipment to be used, complete with date of purchase, model and serial numbers, calibration records (include procedure, source of the standard, last three calibration dates, recommended interval, and curves), and application restrictions.
6. Samples of field reports, charts, and forms proposed to document measured field conditions.
7. A cost proposal to produce a computerized statistical analysis of the field tests, delivered on floppy disks compatible with the Owner's management information system. This analysis shall include averages, standard deviation, ranges, and distribution curves that will permit database trend analysis. The Owner is prepared to provide licensed software to develop the database.

B. The following written progress reports shall be furnished to the director at the earliest possible time after completion of work in each area:
   1. Preliminary (draft) field reports compiled from each of the certification steps.
   2. Copy of the working field logs for the Owner's review and evaluation assembled and distributed at the end of each week or at the completion of each specific test procedure or at the finish of all tests in a specific area or cleanroom module.
   3. Evaluation of any problems which jeopardize final certification results or schedule.
   4. Draft copy of the final assembled as-built certification report.

C. The final completed as-built certification report shall be furnished within 30 days after completion of field activities and include the following documentation at a minimum:

1. Typed or computerized field reports, charts, and forms complete with measured data referenced to sample location.
2. Written description of operating condition of each clean area.
3. Reduced set of architectural floor plan drawings, maximum size 11 by 17 inches (or metric equivalent), made from the project AutoCAD Contract Documents, obtained on floppy disks from the Engineer (after review and clearance by the Owner), showing all test and sample locations referred to on other field data sheets, as well as test results of final field conditions.
4. Separate section in the report outlining any operating or contamination problems remaining at the end of the testing and certifying procedures. Describe the condition and its effect on cleanroom performance.
5. A list of instrumentation and test equipment actually used in the certifying process, including manufacturer, model and serial numbers, and last calibration date.
6. Written description of tests performed, including the purpose, instrumentation, procedure, results, and analysis of the data. Data shall be properly presented and graphically displayed to permit full understanding of the tests. Include the date tests were taken and the names of field technicians performing the tests.
7. One hard-bound copy and one loose, unbound reproducible copy of the completed certification report submitted for the Owner’s review and acceptance.
8. Electronic copies of data and reports that were assembled by the CCC and stored on floppy computer disks.
9. The entire certification report, including charts, drawings, tables, and graphs, assembled and stored in electronic media on floppy computer disks. Drawing files shall be in DXF format, and word files shall be in either MS-DOS, WordPerfect, Lotus 1-2-3, MS Word, or MS Excel format.
10. An analysis with recommendations relating to test results and operating conditions of each area tested.
11. A statement that certification work was performed in accordance with the Contract Documents.

D. A copy of the report shall be kept by the CCC for a minimum of 7 years and shall be available for examination, reproduction, discussion, or clarification during that time.

PART 2 -- PRODUCTS

2.1 MATERIALS

A. The CCC shall supply necessary personnel, materials, tools, test equipment, aerosol generators, instrumentation, and computers required to perform and analyze the cleanroom certification procedures described in this Section.

B. Cleanroom garments and accessories will be furnished and laundered by the Owner.

C. Once the cleanroom ceiling filters have been installed, DOP aerosol shall not be used for challenge of HEPA/ULPA filters. Filter challenges shall be accomplished with polystyrene latex (PSL) spheres only.

D. The Owner will provide suitable office space with telephone and reproducing machines and also a secure/locked storage area for test equipment.

2.2 INSTRUMENTATION CONSIGNED TO THE OWNER

A. The CCC may be authorized to procure certain special test equipment which might be used for specific cleanroom test procedures, then be transferred to the Owner.

B. The equipment shall enhance the Owner’s quality assurance program with the following features:
   1. Upgrade measurement equipment inventory.
   2. Provide reference measurement equipment for initial as-built certification, as well as subsequent tool hookup certification and ongoing operating profiles.
   3. Permit direct correlation between recorded test data accumulated throughout the life of the cleanroom.
   4. Permit calibration of fixed particle measurement analyzers installed for process quality management.

C. At the end of the Contract period, the CCC shall return all used equipment to an authorized service agent for any necessary repairs and for NIST recalibration. The CCC shall bear all costs of repair and/or replacement for damage incurred during the Contract period. This equipment shall be transferred without additional cost directly to the Owner for ongoing cleanroom monitoring.

D. The CCC shall include in the proposal the cost to purchase one of each of the following measurement equipment items which are to be transferred to the Owner. Should new or enhanced equipment be available at the beginning of certification, the Owner retains the right to negotiate a value-added upgrade. Submit separate pricing for each product, including the accessories specified.

E. Laser Particle Counter:
1. Acceptable Manufacturer: Particle Measuring Systems (PMS), Model LPC-0710.
2. Description:
a. Optical laser light-scattering particle counter rated on 0.07- to 1-micron particle size sensitivity with equal to or greater than 50 percent counting efficiency and 5 percent maximum sizing error on a 0.1-micron range at sample flow rate of 1 cubic foot per minute. Sample air discharged from the counter shall be HEPA filtered to add not more than one particle per cubic foot sized 0.1 micron.
b. Counter shall incorporate data storage and statistical calculation capability, remote data output, front panel control and readout, eight-channel measurement display for 0.07- to 1-micron particle size, thermal tape printer with cleanroom data paper, internal filtered vacuum pump, isokinetic sample probes, handheld probes, room sample probes, temperature and humidity environmental sensors, stainless steel cart, and storage/carrying case.

F. Condensation Nucleus Counter:
1. Acceptable Manufacturer: TSI, Model 7610.
2. Description:
a. Ultrafine condensation nucleus counter (CNC) rated at 0.01- to 0.1-micron particle size sensitivity on a sample flow rate of 0.1 cubic foot per minute and 90 percent counting efficiency for 0.02-micron range sample.
b. Counter shall incorporate RS 7130 remote data processor, front panel control and readout, thermal printer with cleanroom data paper, interconnect hardware, internal filtered vacuum pump, stand, room sample probe, and storage/carrying case.

G. Condensation Nucleus Counter:
1. Acceptable Manufacturer: HIAC/Royco High-Flow CNC.
2. Description:
a. High flow condensation nucleus counter (CNC) rated at 0.01-micron particle size sensitivity on a sample flow rate of 1 cubic foot per minute and 90 percent counting efficiency for 0.01-micron range sample.
b. Counter shall incorporate integral data processor, front panel control and readout, thermal printer with cleanroom data paper, interconnect hardware,

internal filtered vacuum pump, wheeled stand, room sample probe, bar code tag reader, and storage/carrying case.

H. Electronic Microanemometer:
   1. Acceptable Manufacturer: Shortridge Instruments AirData Multimeter, Model ADM-870.
   2. Description:
      a. Electronic microanemometer rated at plus or minus 3 percent of airflow and velocity readings and plus or minus 2 percent of pressure readings with sea level air density correction.
      b. Package shall include microprocessor with memory and back-pressure compensation functions, 24- by 24-inch, 24- by 36-inch, and 24- by 48-inch flow hoods, Velgrid, Velprobe, and Temprobe sample nozzle, grid, test stand, battery pack with charger, and storage/carrying case.

I. Portable Laser Particle Counter:
   1. Acceptable Manufacturer: Met-One, Model 227.
   2. Description:
      a. Portable handheld laser particle counter rated on 0.3- to 5-micron particle size on a sample flow rate of 0.1 cubic foot per minute.
      b. Counter shall incorporate rechargeable battery pack and charger, internal memory to record particle counts and size distribution, compatible computer taps to print data on remote printer or computer, integral isokinetic probe, purge filter, and storage/carrying case.

J. Surface Particle Counter:
   1. Acceptable Manufacturer: Dryden, Model Quantum QIII.
   2. Description:
      a. Surface particle detector rated on 0.1-micron and larger particle sizes on a sample flow rate of 1 cubic foot per minute.
      b. Counter shall incorporate external printer, cleanroom printer paper, HEPA discharge filter, and storage/carrying case.

K. Portable Light Meter:
   1. Acceptable Manufacturer: Minolta, Model T-1.
   2. Description: Digital handheld illumination level meter shall include rechargeable battery pack, charger, and storage/carrying case.

L. Portable Electrostatic Meter:
   1. Acceptable Manufacturer: Nilstat 775.
   2. Description: Digital handheld electrostatic field meter shall include chargeplate, rechargeable battery, charger, and storage/carrying case.

2.3 INSTRUMENTATION FOR CERTIFICATION

A. Bench Test Assembly:
  1. Acceptable Manufacturers:
     a. PMS, Model LPC-0710, or Turbo Lasair 110.
     b. Met-One, Model A2120, Model A2090 Plus, or Model 2100 Plus.
  2. Description:
     a. Test stand for bench-scanned ceiling HEPA/ULPA filter media leakage test shall include flow test bench complete with filter face edge seals, frame clamps, PSL generator, nozzle distribution manifold, and discharge encapsulation shroud.
     b. Particle counter for bench test shall be optical laser particle counter rated on 0.07- to 1-micron particle size sensitivity with equal to or greater than 50 percent counting efficiency and 5 percent maximum sizing error on a 0.1-micron range at sample flow rate of 1 cubic foot per minute.
     c. Counter shall incorporate front panel control with readout, remote data transmission, thermal tape printer with cleanroom paper, stainless steel stand, room and isokinetic sample probes, and storage/carrying case.
     d. PSL spheres and generator shall produce a monodispersed challenge of 0.2-, 0.25-, or 0.3-micron-diameter particle with 90 percent of the material to be of the size designated 0.25 micron. The average upstream concentration of the challenge shall be one times 10^6 (1,000,000) particles per cubic foot with a minimum challenge of nine times 10^5 (900,000) particles per cubic foot injected at all times.

B. Laser Particle Counter:
  1. Acceptable Manufacturers:
     a. PMS, Model LPC-0710, or Turbo Lasair 110. An alternative PMS Model μLPC-110 particle counter will be acceptable for this test procedure.
     b. Met-One, Model A2120, Model A2090 Plus, or Model 2100 Plus.
  2. Description:
     a. Particle counter for ceiling filter leakage scan shall be optical laser particle counter rated on 0.07- to 1-micron particle size sensitivity with equal to or greater than 50 percent counting efficiency and 5 percent maximum sizing error on a 0.1-micron range at sample flow rate of 1 cubic foot per minute. Counter must zero count.
     b. Counter shall incorporate front panel control with readout, remote data transmission, thermal

tape printer with cleanroom paper, stainless steel stand, room and isokinetic sample probes, and storage/carrying case.

C. Laser Particle Counter:
1. Acceptable Manufacturers:
   a. PMS, Model LPC-0710, or Turbo Lasair 110.
   b. Met-One, Model A2120, Model A2090 Plus, or Model 2100 Plus.
2. Description: Particle counters for airborne particle count tests shall be optical laser particle counters as specified in paragraph 2.3.B above.

D. Condensation Nucleus Counter:
1. Acceptable Manufacturers:
   a. TSI, Model 7610.
   b. Met-One, Model CNC-1104.
2. Description: Particle counter for special baseline airborne particle count tests shall be the CNC counter scheduled for transfer to the Owner as specified in paragraph 2.2.F above.

E. Electronic Microanemometer:
1. Acceptable Manufacturer: Shortridge Instruments AirData Multimeter, Model ADM-870.
2. Description:
   a. Electronic microanemometer for airflow volume and velocity test shall be rated at plus or minus 3 percent of airflow and velocity readings and plus or minus 2 percent of pressure readings with sea level air density correction.
   b. Package shall include microprocessor with memory and back-pressure compensation functions, 24- by 24-inch, 24- by 36-inch, and 24- by 48-inch flow hoods, Velgrid, Velprobe, and Temprobe sample nozzle, grid, test stand, battery pack with charger, and storage/carrying case.

F. Airflow Test Stand:
1. Acceptable Manufacturer: CCC furnished.
2. Description:
   a. Ribbon test stand for airflow parallelism tests shall include 8- by 8-foot portable stand with slim-profile or aerodynamic top frame set at a distance 12 inches below the filter face grid, fitted with strips of ribbon streamers suspended 24 inches on center along the top frame and reaching to within 24 inches of the floor. Streamers shall be dark-colored nonshed monofilament polyester ribbon similar to Flo-Viz or 0.05-mil polyethylene ribbon.
   b. Test stand shall include two suspended concentric rings arranged 12 inches below the ceiling and

above the floor, sized to outline an imaginary cone with sloping sides to encompass the air path specified.
   c. Frame shall be dismantled for access between rooms.

G. Electronic Microanemometer:
   1. Acceptable Manufacturer: Shortridge Instruments AirData Multimeter, Model ADM-870.
   2. Description: Manometer for pressurization tests shall be microanemometer as specified in paragraph 2.3.E above.

H. Laser Particle Counter:
   1. Acceptable Manufacturers:
      a. PMS, Model LPC-0710, or Turbo Lasair 110.
      b. Met-One, Model A2120, Model A2090 Plus, or Model 2100 Plus.
   2. Description: Particle counters for enclosure induction leak tests shall be optical laser particle counters as specified in paragraph 2.3.A above. An alternative PMS Model μLPC-110 particle counter will be acceptable for this test procedure.

I. Recovery Test Stand:
   1. Acceptable Manufacturer: Dryden, Model DE 342 CRF.
   2. Description: DryFog ultrapure deionized fogger for recovery rate tests shall include wheeled cart, distribution hose and nozzles, stainless steel holding tank for deionized water, and portable two-speed fan compatible with Class M1 criteria.

J. Laser Particle Counter:
   1. Acceptable Manufacturers:
      a. PMS, Model LPC-0710, or Turbo Lasair 110.
      b. Met-One, Model A2120, Model A2090 Plus, or Model 2100 Plus.
   2. Description: Particle counter for makeup air handler HEPA/ULPA final filter media tests shall be optical laser particle counters as specified in paragraph 2.3.A above.

K. Electronic Temperature and Humidity Data Logger:
   1. Acceptable Manufacturers:
      a. PMS Multiport, Model HHIPA-1.
      b. Vaisala, Model HMI-31/HMP-111A.
   2. Description:
      a. Temperature and humidity sensor for environmental uniformity/stability tests shall be multipoint electronic recorder and data logging instrument. Electronic temperature sensor shall be CAI, Model TE 300, sensitive to 0.01 degree F resolution and accurate to 0.1 degree F dry bulb.

Dielectric thin-film capacitor dew point temperature sensor shall include a data logger sensitive to plus or minus 0.1 percent resolution and accurate to 0.1 degree F dew point.

b. Sensor shall incorporate remote data transmission, front panel control and readout, thermal tape printer with cleanroom paper, stand, and storage/carrying case.

L. Sound Meter:
1. Acceptable Manufacturers:
    a. Bruel and Kjaer, Model 2230.
    b. General Radio, Model 1982.
    c. Scott Instruments, Model 451.
    d. Larson-Davis Labs, Model 800B.
2. Description:
    a. Sound meter for operating noise level tests shall be an eight-octave band analyzer with octave band filter and sound level calibrator conforming to ANSI S1.4-1983 for sound meters and S.11-1986 for noise filters.
    b. Meter shall incorporate thermal tape printer with cleanroom paper, pickup microphone sensors, and storage/carrying case.

M. Vibration Accelerometer:
1. Acceptable Manufacturers:
    a. Bruel and Kjaer, Model 4379.
    b. WaveTek Inc., Model 5810.
2. Description:
    a. Vibration accelerometer for floor vibration test shall be 400-line Fourier transform analyzer accurate to 1 dB or plus or minus 10 percent.
    b. Instruments shall include digital plotter, seismic accelerometer, amplifier with 1/3-octave band analyzer, and FFT analyzer.

N. Portable Light Meter:
1. Acceptable Manufacturers:
    a. Minolta, Model T-1.
    b. Simpson, Model 408, or M/N 441.
    c. General Electric, Model 214.
    d. Silvania, Model DS 2000.
2. Description: Light meter for lighting level test shall be illumination level meter rated between zero and 500 foot-candles with storage/carrying case.

O. Light Wave Meter:
1. Acceptable Manufacturers:
    a. WaveTek, Model 580-A.
    b. UVP, Model UVX-25.
    c. International Light, Model IL-1700.

2. Description:
    a. Light wave meter for wavelength spectrum test shall be complete with digital plotter, test probes, and storage/carrying case.

P. Conductance Meter:
1. Acceptable Manufacturers:
    a. AEMC, Model 1100.
    b. Monroe Electronics, Model 278.
    c. Biddle, Model 212559.
2. Description: Conductance meter for floor conductivity tests shall include floor test kit rated at open-circuit voltage of 500 volts dc and scale range of 10K ohms to 50 megohms with storage/carrying case.

Q. Portable Electrostatic Meter:
1. Acceptable Manufacturer: Nilstat 775.
2. Description: Ion meter for electric space charge test shall be the handheld electrostatic field meter scheduled for transfer to the Owner. Meter was specified in paragraph 2.2.L above.

R. Magnetic Field Test Kit:
1. Acceptable Manufacturers:
    a. Perkin-Elmer magnetic field test kit.
    b. Magnetic Sciences, Model 20.
    c. Walker Scientific, Model CLF-80D.
    d. Lake Shore Cryotronics, Model 450.
2. Description: Magnetic field search coil for electromagnetic interference test shall include search coil, data physics line Fourier analyzer, signal conditioner, digital plotter, meter, and storage/carrying case.

S. Particle Fallout Test Kit:
1. Acceptable Manufacturer: CCC furnished.
2. Description: Witness test plate for total particle fallout test shall be 4-inch-diameter glass petri dish or silicon wafer prepared with gel coating consisting of one part Vaseline per nine parts solvent, then baked in 85 degree C oven for 10 minutes. Samples will be analyzed by the Owner.

T. Air Particle Analyzer:
1. Acceptable Manufacturers:
    a. Coulter LS Series.
    b. California Measurements, Model MPS-4G1.
2. Description: Air separator for particle identification test shall be air particle analyzer, including vacuum pump, four-stage particle separator, SEM mounting plates, and cart. Samples will be analyzed by the Owner.

U. Hydrocarbon Analyzer:
  1. Acceptable Manufacturers: Hewlett Packard 9000 Series 300 computer, Hewlett Packard Model 5890 gas chromatograph, and Model 5971A autosampler.
  2. Description: Air analyzer for hydrocarbon/offgas elemental test shall be charcoal sample tubes, DuPont Alpha-1 calibrated air sampler pump, flame ionization detector, and carbon disulfide desorber chemicals.

V. Nomura Elemental Test Kit:
  1. Acceptable Manufacturer: CCC furnished.
  2. Description: Collection container for Nomura Technique elemental mass spectrometry test shall be a 4-inch-diameter jar filled with ultrapure 18-megohm deionized water. Samples will be analyzed by the Owner.

W. Viable Particle Test Kit:
  1. Acceptable Manufacturer: CCC furnished.
  2. Description: Witness test plate for viable particle fallout test shall be a 4-inch-diameter glass petri dish prepared with protein-based agar gel. An Anderson or DuPont air sample collector or slit selector cover shall be used to provide selective sample rates. Samples will be analyzed by the Owner.

X. Three-Dimensional Anemometer:
  1. Acceptable Manufacturer: Kaijo Denki, Model WA-390.
  2. Description: Anemometer for three-dimensional airflow turbulence test shall be ultrasonic anemometer with microprocessor, three-directional flow sample nozzle head, software, test stand, and storage/carrying case.

Y. Sodium Test Kit:
  1. Acceptable Manufacturer: CCC furnished.
  2. Description: Air analyzer for sodium test shall be DuPont Alpha-1 calibrated air sampler pump, 100-ml Nalgene polypropylene or quartz impinger with cap and sample tubes, DI water, and 0.5-micrometer dry Teflon filter strainers. Samples will be analyzed by the Owner.

PART 3 -- EXECUTION

3.1 REQUIRED SITE CONDITIONS

A. Facility Completion at Protocol Stage 4:
  1. Building perimeter walls, roof, and accessories shall be installed to create a pressurized envelope around the Clean Zone.
  2. Clean Zone perimeter walls, ceiling, raised floor panels, doors, and any necessary interior partitions

shall be installed that are essential to successful system performance. If approved by the director, temporary barriers may be used for area isolation.
3. Permanent personnel gowning area shall be in operation.
4. Final wipedown cleaning procedures shall be complete on:
    a. Cleanroom finished surfaces.
    b. HVAC system ducts, plenums, and air handler surfaces exposed to airflow.
    c. Wall and floor cavities used as part of the cleanroom air handling strategy.
    d. Building structural elements and utility systems in contact with the cleanroom airstream.
5. Tool hookup or miscellaneous activities shall be curtailed in the test area so equipment heat and contamination loads cannot influence the test results and personnel movement will not alter flow fields.

B. Facility Utility Support Systems:
1. Air-handling systems serving the Clean Zone shall be installed and operating under automatic controls.
2. Initial testing and balancing shall be complete on the air and water systems serving the Clean Zone air-handling systems.
3. Process exhaust systems and pressurization control fans shall be installed and operating to generate simulated exhaust rates from the Clean Zone.
4. Fan vibration and noise tests shall be complete.
5. Cleanroom lights, sprinklers, and safety devices shall be installed and operational.
6. Housekeeping vacuum system shall be operational and used by janitorial crews.
7. Support systems required to perform certification tests shall have been operating normally for a minimum stable period of 5 days.

3.2 CERTIFICATION PREPARATION, SETUP, AND PROTOCOL

A. Activities within the facility shall be in compliance with the requirements of Section 01355, Clean Zone General Requirements, and shall be performed under the direction and coordination of the provisions of Section 13004, Cleanroom Zone Systems.

B. All CCC's field staff shall complete the following special tasks prior to performing any certification activities:
1. Attend one protocol training class taught by the director.
2. Attend one hazardous material safety training class taught by the Owner, then satisfactorily pass the exam.

3. Obtain color-coded security badges.
4. Obtain approved cleanroom footwear.
5. Complete field correlation of cleanroom measurement instruments to be used by other validation groups.

C. The CCC shall inspect the entire Clean Zone, accompanied by the director, and highlight any existing conditions that could jeopardize the certification results. No certification steps shall proceed without the director's written release, and a copy of the director's written authorization shall be furnished to the Owner’s contamination control officer.

D. The CCC shall coordinate field certification activities with the director and the Owner’s contamination control officer to permit observation of any test procedure.

E. Certification activities shall be performed in increments of one cleanroom element, one clean aisle with adjacent cores, or building structural module, unless scheduled otherwise by the director.

F. Any filters removed and/or packed and stored will be handled by the Contractor.

G. The CCC bid shall allow for retest as stated within each test procedure.

3.3 CERTIFICATION PROCEDURES AND FIELD TESTS

A. Bench-Scanned Ceiling HEPA/ULPA Filter Leakage Test:
   1. Purpose of Test:
      a. Determine integrity of ceiling HEPA/ULPA filter modules showing signs of shipping/handling damage before installation into ceiling grid.
      b. Determine integrity of each truckload of filter shipments by spot testing random statistical samples.
      c. Determine integrity of HEPA/ULPA filter modules before installation into makeup air handler filter racks.
      d. Test Procedure: Test every fifth filter module which shall determine the fate of the previous four modules.
      e. Provide flow test bench and optional laser particle counter.
      f. Introduce PSL challenge spheres from the generator into the inlet of the air blower. A challenge of at least two times 10^7 particles per cubic foot sized 0.2 to 0.26 micrometer is required at the filter.
      g. Test the upstream challenge when each filter module is placed in the bench.

h. Scan the entire downstream filter face area in overlapping strokes, moving the probe at 2 inches per second, spaced a distance of 1 inch from the filter face.
i. Visually inspect the urethane media potting sealant on both the upstream and downstream sides at the media perimeter for cracks or gaps in the sealant.
j. Retest repaired filter modules judged acceptable by the director.

2. Acceptance Criteria:
   a. Reject filter modules with direct leaks detected at the perimeter frame, center divider, center test ports, or within the media pack.
   b. Reject any filter module when aerosol penetration exceeds 0.005 percent of the upstream challenge concentration.
   c. Reject any filter module with visible cracks in either the upstream or downstream beads of media edge potting sealant.
   d. Reject or accept the test filter and the previous four modules where batch sampling technique is used.
3. Documentation:
   a. List the model and serial number of rejected filters.
   b. Identify the nature/location of the leak, the test challenge size/concentration, and the measured penetration/particle count.
   c. Create statistical data showing filters tested with acceptance and rejection rate.

B. Installed Ceiling HEPA/ULPA Filter Leakage Scan Test:
   1. Purpose of Test:
      a. Determine integrity of ceiling HEPA/ULPA filter modules after installation in the ceiling grid.
      b. Determine leakage through any component in the ceiling assembly.
      c. Locate leaks created by ceiling defects.
      d. Determine HEPA/ULPA filter media shedding.
   2. Test Procedure:
      a. Test the entire ceiling assembly throughout every cleanroom as identified in Table T-2 after completion of the installation.
      b. Provide optical laser particle counters, PSL aerosol generator, and accessories. Automatic filter scanning equipment will not be acceptable.
      c. Introduce PSL spheres from the generator into the inlet of the of each fan filter unit individualy ducted filter. An average challenge of 10^6 particles per cubic foot sized 0.2 to 0.3 micron is required at the filter, although a challenge of two times 10^6 is desired.

d. Measure and record the upstream challenge at the filter at least once every hour. Measure upstream filter challenge at two test ports in each packaged ceiling plenum module.
e. Scan the entire downstream filter face area isokinetically in overlapping strokes, moving the probe at 2 inches per second or 1 square foot per minute, at a distance 1 inch below the filter face. Scan the entire perimeter, center support mullions, and corners at a traverse rate of 10 fpm maximum. Provide side shields for filter modules located in partial-coverage ceilings to block induced air influx from adjacent blank panels. Bidders' work scope shall allow 10 minutes scan time per filter module, including all joints.
f. Scan joints in the ceiling assembly, including the gap between the ceiling grid and filter module, wall-to-ceiling joint, sprinkler pipe and electrical conduit penetrations, and blank panel edge seals. Blank panels do not require scan tests where the ULPA filters are the ducted style.
g. Extend test time under any portion of the filter showing a single leak by sampling directly under the leak. A reading of 150 counts in a 10-second time span constitutes a significant leak requiring repair.
h. Scan each joint and the entire area encompassed by the side shrouds under the ceiling automated wafer transporter.
i. Document in the logbook the filter serial number and failure criteria for rejected filter modules. Repairs and leaks will be corrected by others.
j. Retest those repaired filter modules judged acceptable by the director. Assume retest of 5 percent of the total filters in the ceiling assembly measured during initial tests.
k. Repeat the filter scan test for 10 percent of installed ceiling filters without any test challenge to determine particles shed by the media itself. Perform scan under media face only as described in paragraph B.2.e above.

3. Acceptance Criteria:
a. Reject filter modules with direct leaks detected at the perimeter frame, center divider, center test ports, or within the media pack.
b. Reject any filter module when aerosol penetration exceeds 0.1 percent of the upstream challenge concentration.
c. Locate and tag leaks through the ceiling assembly that require sealing.
d. Reject any filter module with more than ten individual repairs or with a single repaired area

of 1 square inch or 1 linear inch or with an accumulative total repair area of 1 percent of media pleated face area.

4. Documentation:
   a. List the model and serial number and installed location of rejected filters, including failure criteria.
   b. Identify the nature/location of the leak, the test challenge size/concentration, and the measured penetration/particle count. Particle size penetration/shedding shall be recorded 0.1, 0.2, 0.3, 0.5, 1, and 5 microns.
   c. Record location of filter modules or components that failed, mapped on a cleanroom reflected ceiling layout.
   d. Identify repaired leaks that were accepted by the director.
   e. Create and maintain a logbook showing test procedures and results. Logbook shall be maintained by the CCC, then verified by the director at the end of each day.

C. Airborne Particle Count Test:
1. Purpose of Test: Determine particle count levels at reference particle size to ensure systems operate within design criteria.
2. Test Procedure:
   a. Measure and record the airborne particle counts throughout every cleanroom of any class as identified in Table T-2.
   b. Complete the installed ceiling HEPA/ULPA filter leakage scan test, makeup air handler HEPA/ULPA final filter media test, parallelism test, pressurization test, air velocity uniformity test, and the enclosure induction leak test before starting airborne particle count sampling.
   c. Measure and record the particle count at a distance 42 inches above the floor in the center of each test sector or 12 inches above any process equipment obstructing the standard test station. Allow ample time span and sample volume as stated in Federal Standard 209 with sector size as follows:
      1) Sector 4 feet by 4 feet in Class M1.5 through M3.5 (100) with 4-cubic-foot sample volume.
      2) Sector 16 feet by 16 feet in Class M4.5 (1,000) areas with 1-cubic-foot sample volume.
      3) Sector 48 feet by 48 feet in Class M5.5 (10,000) areas.
      4) Sector 10 feet by 10 feet in un-rated equipment/service chases.
      5) Sector 20 feet by 20 feet in un-rated interstitial return air plenums.

d. Perform at least one measurement in every rated cleanroom smaller than the sector test area.
e. The CCC may use the sequential sample counting procedures described in Federal Standard 209. If this system is employed, the procedure shall be described in the Bid.
f. Measure and record particle concentration 3 feet outside each cleanroom door opening into an area less clean (higher contamination class) for a period of 2 minutes each. The door shall be in the closed position.
g. Measure and record particle concentration inside each cleanroom chemical/parts pass-through cabinet for a period of 2 minutes each. For purposes of the proposal, perform tests in pass-through cabinets.
h. Measure and record particle concentration at two fixed stations under the ceiling automated wafer transporter as the transport carriage moves over the sample probe a total of ten times.
i. Identify, tag, and record any test stations or sectors that do not meet design criteria and require corrective action.
j. Retest those test stations or sectors that require corrective modifications in order to achieve design criteria. Assume retest of 5 percent of the total test stations measured during initial tests.

3. Acceptance Criteria:
    a. Obtain airborne particle counts necessary to determine the room class identified in Table T-3.
    b. Compute the following data to establish compliance with these acceptance criteria for each room classification:
        1) The particle concentration measured at each station falls at or below the values listed in the table.
        2) Particle concentrations measured underneath the automated wafer transporter does not exceed a value of 100 times the average room background of adjacent measurements.
        3) If sample size is less than ten readings per clean area, calculate 95 percent upper confidence limit, and this value shall fall below the specified area class limits.
        4) If sequential sampling is used, sample techniques shall follow Appendix F of Federal Standard 209 criteria.
4. Documentation:
    a. Record particle counts for sizes 0.1, 0.2, 0.3, 0.5, and 1 micrometer in each test station mapped on a cleanroom layout, although only the

0.3-micron reference size shall be used to establish room acceptance.

b. Create bar chart showing particle counts versus frequency/repeatability count, as well as minimum, mean, and maximum values.

c. Copy one representative strip chart from each clean aisle/test station printed during the test procedures.

D. Baseline Airborne Particle Count Test (Alternative 1A1):

1. Purpose of Test: Create baseline particle count levels on reference size 0.02 micron at strategic locations within the cleanroom using test equipment being consigned to the Owner.

2. Test Procedure:

a. Measure and record particle counts for 10 minutes each at 30 test stations selected by the Owner.

b. Complete the airborne particle count test and prerequisite tests before starting benchmark particle count sampling.

c. Obtain the particle counts at a distance 42 inches above the floor using both the optical and CNC particle counters at the same test stations and interval.

d. Sample for particle count at each test station as described for time interval and sample volume in accordance with Federal Standard 209.

3. Acceptance Criteria:

a. Obtain airborne particle counts necessary to determine the room class identified in Table T-3.

b. Compute the following data for each area: the average of particle counts measured at each test station to determine correlation with Table T-3; the mean of these averages.

4. Documentation:

a. Record particle counts for sizes 0.01, 0.02, 0.05, 0.1, 0.2, 0.3, and 0.5 micron at each test station mapped on a cleanroom layout.

b. Create bar chart showing particle counts versus frequency/repeatability count.

c. Create graphs showing particle counts versus time for the 10-minute test interval at each test station.

d. Copy strip charts printed during the test procedures.

E. Airflow Volume and Velocity Test:

1. Purpose of Test:

a. Determine supply airflow volume delivered through each ceiling filter.

b. Determine volume uniformity throughout the clean aisle/cleanroom.

c. Determine air velocity profile 12 inches below the face screen of each ceiling filter.

2. Test Procedure:
   a. Document the supply airflow volume for any filter module that falls outside the tolerance required for acceptance. Air adjustments will be performed by the TABC.
   b. Measure and record the air velocity profile for 10 seconds each at 30 test stations selected by the Owner. Two Velgrid readings are required below all size 2- by 4-foot filter modules. Select the average of the two values for the recorded reading.
3. Acceptance Criteria:
   a. Obtain supply airflow volume for every ceiling filter module to fall within plus or minus 10 percent of the design air supply volume
   b. Obtain air velocity readings at a distance 12 inches below the plane of the ceiling [to create initial baseline reference velocity profile at certification] [that falls within plus or minus 20 percent of the mean of all velocity values from that respective filter module only if the filter module has been fitted with airflow adjustment damper].
4. Documentation:
   a. Record air supply volume for each filter module mapped on a cleanroom layout.
   b. Record the low, average, and high air velocity readings for each filter module mapped on a cleanroom layout.
   c. Compute and document cleanroom average airflow volume from data collected for each filter module.

F. Airflow Parallelism Test:
1. Purpose of Test:
   a. Verify parallel vertical flow paths of supply airflow.
   b. Detect and eliminate horizontal and vertical upflow air patterns.
2. Test Procedure:
   a. Measure and record the parallel vertical flow path throughout every clean aisle/cleanroom of Class M1.5.
   b. Make provisions in the clean aisle/cleanroom to set the test stand in 8- by 8-foot test sectors. Where clean aisle is 10 feet wide or narrower, take two rows of readings.
   c. Observe and photograph vertical air path at 20 critical test stations selected by the Owner.
   d. Document changes in vertical air paths near each door as it opens and closes.

e. Document areas where air deflection exceeds acceptance criteria.
f. In lieu of the test procedure described, the CCC is authorized to use strips of the monofilament polyester ribbon taped to the ceiling grid runners or light lens on 4-foot centers. Ribbon length shall be suitable so the bottom end is within 24 inches of the floor.
g. Retest those test sectors that require corrective modifications or system air balancing in order to achieve design criteria. Assume retest of 5 percent of the total test stations measured during initial tests.

3. Acceptance Criteria:
   a. Achieve parallel vertical flow paths whereby 100 percent of all test stations fall within 30 degrees of vertical, and 80 percent of all test stations fall within 20 degrees of vertical in any two perpendicular room axes for all areas with standard undampered floor panels, solid floor panels, or low-wall return grilles.
   b. Confirm absence of vertical upflow air patterns anywhere in the cleanroom.
4. Documentation:
   a. Record nonconforming air patterns in any test sector as a velocity vector mapped on a cleanroom layout.
   b. Record, calculate, and document the deflection of air patterns mapped on a cleanroom layout.
   c. Provide 35-mm photographs of air path at the 20 selected test stations.
   d. Identify probable cause of any test sectors where vertical path cannot be maintained and recommend corrective action.

G. Pressurization Test:
1. Purpose of Test:
   a. Confirm capability of cleanroom air-handling systems to maintain cascaded air pressure regime within the Clean Zone.
   b. Verify performance of pressure controls as doors are activated.
   c. Confirm pressure differential across return air components.
2. Test Procedure:
   a. Measure and record the relative pressurization for every clean aisle/cleanroom of Class M1.5 through M4.5 .
   b. Measure and record the relative positive pressure between each clean aisle/cleanroom and the adjacent area of any cleanliness class physically separated by a wall.

c. Measure and record the relative positive pressure between each clean aisle/cleanroom and adjacent areas of lesser class when just one of the interconnecting doors is open.
d. Measure and record the relative positive pressure sequentially from the area with the highest cleanliness requirement outward through contiguous spaces to the outdoors. Verify the pressurization cascade identified under cleanroom criteria in paragraph 1.6.A.
e. Measure and record the relative positive pressure across every pass-through, vestibule, and air lock.
f. Measure and record the differential pressure across return air registers, perforated and grated floor tiles, and relief air dampers at 20 test stations selected by the Owner.
g. Retest those zones that require system air balancing in order to achieve design criteria. Assume retest of 5 percent of the total test stations measured during initial tests.

3. Acceptance Criteria:
a. Achieve continuous positive pressure gradient from the clean aisle/cleanroom to adjacent areas in compliance with design criteria when all doors are in the normally closed position. Minimum acceptable pressure differential shall be set at 0.01 inch WC positive.
b. Maintain positive pressure from the clean aisle/cleanroom to adjacent areas of higher contamination with one interconnecting door open.

4. Documentation:
a. Record relative room pressure differentials and overall pressurization

H. Installed Makeup Air Handler HEPA/ULPA Final Filter Media Test:

1. Purpose of Test:
a. Determine integrity of HEPA/ULPA filter modules after installation in the makeup air handler final filter frame.
b. Determine particle contribution from the duct distribution system to the makeup airstream.

2. Test Procedure:
a. Test the entire final filter assembly in every makeup air handler after completion of the installation when the units are operating.
b. Provide optical laser particle counter, PSL generator, isokinetic sample probes, and accessories.
c. Introduce PSL spheres from the generator into the [outside air intake plenum of the air handler] [upstream plenum of the final filter assembly].

A challenge of greater than 10^6 particles per cubic foot sized 0.2 to 0.26 micrometer is required at the generator outlet.

d. Measure and record the upstream challenge at the final HEPA/ULPA filter.
e. Measure particle counts in the airstream downstream of the HEPA/ULPA final filter.
f. Scan all joints, perimeter edge seals, and corners in the final filter frame assembly.
g. Extend test time for any portion of the filter showing high particle counts of at least 150 particles in a 10-second time span.
h. Identify rejected filter modules for the filter manufacturer to claim and repair. Document the serial number and failure criteria in the logbook.
i. Retest those repaired filter modules judged acceptable by the director. Assume retest of one air handler measured during initial tests.
j. Measure particle counts in the makeup supply airstream immediately upstream of two recirculation air handlers served by each of the makeup air handlers.

3. Acceptance Criteria:
   a. Reject any filter module with direct leaks detected at the perimeter frame, center divider, center test ports, or within the media pack.
   b. Reject any filter bank when aerosol penetration exceeds 0.01 percent of the upstream challenge concentration.
   c. Reject any filter with more than two repairs or repairs larger than 1 square inch each.
4. Documentation:
   a. List the model and serial number of all rejected filters.
   b. Identify the nature/location of the leak, the test challenge size/concentration, and the measured penetration/particle count.
   c. Create statistical data showing filters tested with acceptance and rejection rate.

I. Environmental Uniformity/Stability/Recovery Test:
   1. Purpose of Test:
      a. Confirm the capability of the facility support systems to control temperature and relative humidity to meet the project criteria.
      b. Verify uniformity of environmental conditions throughout contiguous areas of the Clean Zone.
      c. Confirm stability of environmental conditions at control sensing points.
      d. Determine ability of facility support systems to recover to design environmental conditions after an internal thermal upset.

e. Demonstrate ability of the cleanroom air-handling and control systems to achieve a continuous 7-day environmental profile in conformance with project design criteria.
f. Correlate test instrumentation with built-in facility monitoring system equipment.

2. Test Procedure:
a. Measure and record the dry bulb and dew point temperature uniformity/stability for every cleanroom of any class . Environmental tests can be set up and run concurrently with other tests where results are not jeopardized. Support systems shall have been in normal automatic operation under control of calibrated permanent controllers for at least 7 days.
b. Place test probes uniformly within the space with at least one-fourth of the probe sites adjacent to sample points of the FMCS facility monitoring/control system.
c. Position 12 test probes within one clean aisle at a distance 42 inches above the finished floor. For large or open cleanrooms, divide the area into test sectors corresponding to the smoke exhaust with test sectors not to exceed 200 square feet each.
d. Measure and record dry bulb and dew point temperature readings in the first clean aisle/test sector. Samples from the 12 probe sites shall be recorded simultaneously to verify uniformity. Time span for test intervals and overall duration of documentation shall be:

Cleanliness Class	Recording Interval	Recordin g Duration
ISO Class 3	6 minutes	12 hours
ISO Class 4(10) through ISO Class 7 (10,000)	6 minutes	6 hours

e. Relocate test probes to each of the remaining clean aisle/test sectors and repeat the 12-point data recording.
f. Relocate the 12 test probes to one critical clean aisle/test sector selected by the Owner. Repeat the 12-point data recording on 10-second intervals for a 2-hour duration.
g. Adjust temperature and humidity control set points in the same critical clean aisle/test sector to place the space environmental conditions beyond required criteria. From the operating set point, the temperature shall be adjusted upward by 2 degrees F and the dew point

temperature upward by 2 degrees F. The CCC is authorized to use a portable heater and humidifier which can be energized to create internal upset conditions in lieu of adjusting control settings. Repeat the 12-point data recording on a continuous basis for a 2-hour duration during adjustment period.

h. Readjust temperature control set point only back to desired operating condition. Repeat the 12-point data recording on 10-second intervals for a 2-hour duration or until the original set point temperature has been achieved for a 10-minute interval.

i. Readjust humidity control set point only back to desired operating condition. Repeat the 12-point data recording on 10-second intervals for a 2-hour duration or until the original set point dew point temperature has been achieved for a 10-minute interval.

j. Redistribute the 12 test probes so one probe site is located in each of the 12 critical test sectors selected by the Owner from previous clean aisle/test sectors. The probe shall be placed adjacent to the facility monitoring control sample points where possible. Repeat the 12-point data recording on 6-minute recording intervals for a 7-day continuous time span.

k. Retest those clean aisles/test sectors that require corrective modifications or adjustments in order to achieve design criteria. Assume retest of two test station setups measured during initial tests.

3. Acceptance Criteria:

a. Achieve temperature uniformity throughout contiguous areas of the cleanroom from all test station readings within plus or minus 0.5 degree F of the mean temperature value from all readings.

b. Achieve relative humidity uniformity throughout contiguous areas of the cleanroom from all test station readings within plus or minus 5 percent of the mean humidity value from all readings.

c. Demonstrate stability of both temperature and absolute humidity in the 12 critical test sectors to be within the range of uniformity values over a continuous time span of 7 days. Calculate and plot corresponding relative humidity profiles.

d. Verify recovery time is less than 10 minutes for both dry bulb and dew point temperatures to return to normal control set points after an internal upset of both conditions.

4. Documentation:

a. Plot temperature versus time by test station.

b. Plot both dew point temperature and relative humidity versus time by test station mapped on a cleanroom layout.
c. Create charts showing temperature versus frequency/repeatability count by test sector.
d. Create charts showing dew point temperature and relative humidity versus frequency/repeatability count by test sector.
e. Provide data logs and charts for the one critical area selected by the Owner.
f. Provide data logs and charts for the 7-day performance demonstration.
g. Record the average temperature and relative humidity calculated from the 2-hour interval in test sectors, mapped at each test station on a building layout.

J. Noise Level Test:
1. Purpose of Test:
   a. Determine the operating noise level in the cleanroom produced by the facility support systems.
   b. Verify fan performance with manufacturer's submittal data.
2. Test Procedure:
   a. Measure and record the operating noise for every cleanroom of any class M1.5 through M3.5.
   b. Place pickup sensors on a uniformly arranged pattern of 200 square feet per test sector.
   c. Measure and record the sound pressure levels at the center of each test sector at a distance 60 inches above the floor with facility systems operating at normal design performance capacity.
   d. Record sound pressure data in all eight-octave bands and also at a weighted dBA scale.
   e. Retest those test sectors that require corrective modifications or adjustments in order to achieve design criteria. Assume retest of 5 percent of the total test stations measured during initial tests.
3. Acceptance Criteria:
   a. Achieve operating noise levels that do not exceed the noise criteria shown in Table T-2 in all octave bands for a minimum of 90 percent of all test station readings.
   b. Achieve operating noise levels that are 3 dB maximum above the noise criteria shown in Table T-2 in the eight-octave bands for 100 percent of test station readings.
4. Documentation:
   a. Plot sound pressure levels by octave band to establish noise criteria curve for each test sector mapped on a cleanroom layout.

b. Chart dBA readings for each test sector mapped on a cleanroom

K. Lighting Level Test:
   1. Purpose of Test:
      a. Determine the lighting intensity levels and uniformity provided by the normal lighting system in the white-light areas and yellow-light areas identified.
      b. Determine the lighting intensity levels provided by the emergency lighting system in the white-light areas and yellow-light areas identified.
   2. Test Procedure:
      a. Measure and record the white-light lighting levels for every cleanroom of any class as identified in Table T-2. All lighting shall have operated continuously and seasoned for at least 300 hours.
      b. Measure and record the yellow-light lighting levels in all photoprotected yellow cleanrooms identified.
      c. Measure lighting intensity at 42 inches above the finish floor on a nominal 10- by 20-foot test sector.
      d. Measure areas having aisles greater than 20 feet in width:
         1) At approximately the center of the area, take measurements within an area of 10 feet by 20 feet.
         2) At wall parallel to light fixtures, take readings over an area of 10 feet by 20 feet (long dimension parallel to the wall with the first row taken at the wall plane).
         3) At wall perpendicular to light fixtures, take readings over an area of 10 feet by 20 feet (long dimension parallel to the wall with the first row taken at the wall plane).
      e. Measure areas having aisles less than 20 feet in width. Take readings over a 20-foot length of cleanroom aisle.
      f. Measure and record the emergency lighting levels at the same test stations described for normal lighting levels tests.
   3. Acceptance Criteria:
      a. Achieve foot-candle values within plus or minus 20 percent of those indicated in design criteria.
      b. Verify Uniformity of Illumination:
         1) Large Open Areas: 2:1 (average: minimum).
         2) Other Areas: 4:1 (average: minimum).
   4. Documentation:
      a. Record data using a Cartesian coordinate system (X-Y-Z) with the origin (X = 0 and Y = 0) at the southwest corner of the area to be tested.

b. Plot foot-candle values for each test station, mapped on a cleanroom layout. Indicate test stations which had yellow light.
c. Express area results in average maintained foot-candles with average: minimum ratios and standard deviation from design criteria.

L. Light Wave Spectrum Test:
1. Purpose of Test: Verify that luminaires in the photolithography area do not emit light wavelengths that will cause degradation of the photoresist.
2. Test Procedure:
a. Record light spectral output at 20 test stations in the photolithography area selected by the Owner.
b. Monitor each location for 2 minutes.
c. Record output wavelength (manometers) versus radiated power (milliwatts).
d. Take measurements at height of 42 inches above finished floor.
e. Place pre-coated witness test wafers furnished by the Owner at each of the 20 test stations. After 60 minutes of exposure time to room yellow light, examine and record degradation of photoresist. Deliver exposed witness wafers to the Owner for analysis.
3. Acceptance Criteria: Verify no measured light output at wavelengths less than 470 manometers.
4. Documentation:
a. Record the 20 test stations mapped on a cleanroom layout.
b. Provide scaled X-Y plots showing wavelengths (manometers) versus radiated power (milliwatts).
c. Duplicate Owner’s photoresist reactance report from witness wafers.

M. Floor Conductivity Test (Alternative 1A5):
1. Purpose of Test:
a. Determine resistance between specified points on the floor.
b. Determine resistance from the floor covering to building ground at strategic locations within the building.
2. Test Procedure:
a. Perform conductivity tests at 20 test stations selected by the Owner between points on the floor covering in accordance with NFPA 99, Chapter 12-4.1.3.8, except as modified herein and between the floor tile frame and the pedestal bases.
1) Floor shall be tested with temperature and relative humidity maintained within the specified limits and with floor panels

placed/secured normally to the support pedestals.

2) Each electrode (two required) shall weigh 5 pounds and have a flat, circular contact area 2 1/2 inches in diameter, which shall comprise a surface of aluminum or tinfoil 0.0005 inch to 0.001 inch thick, backed by a layer of rubber 1/4 inch thick, and measuring between 40 and 60 durometer hardness as determined with a Shore Type A durometer (ASTM D2240-68).

3) Resistance shall be measured by a suitably calibrated ohmmeter with a tolerance defined as follows:

a) Short-circuit current shall be between 2.5 mA and 5 mA.

b) At any value of connected resistance, Rx, the terminal voltage, V, shall be:

$$\frac{Rx}{Rx + \text{internal resistance}} \quad 500V \pm 15 \text{ percent}$$

c) Internal resistance shall be greater than 100,000 ohms.

b. Measure and record conductivity between points on the floor covering at 20 test stations selected by Owner.

1) Measurements on continuous flooring shall be made between pairs of points with the electrodes 3 feet apart.

2) Measurements on raised floor tiles shall be made between pairs of points:

a) Center of test tile to center of any adjacent tile.

b) Corner of test tile to corner of any tile two positions away.

c) Center of test tile to any supporting pedestal.

d) Center of test tile to conductive paint finish covering main structural concrete floor.

c. Measure and record conductivity from floor covering to building ground (raised floors only) at 20 test stations selected by the Owner. Measurements shall be made between five pairs of points in each room with one electrode on the floor connected to the ohmmeter. The other terminal of the ohmmeter shall be connected to the nearest building column or exposed grounded conductor.

d. Retest those test sectors that require corrective modifications or adjustments in order to achieve design criteria. Assume retest of 5 percent of the total test stations measured during initial tests.

3. Acceptance Criteria:
    a. Verify floor panel-to-panel test achieves average values less than 1 million (10^6) ohms.
    b. Verify floor-to-building ground test achieves average values less than 1 million (10^6) ohms.
4. Documentation:
    a. Record the test stations mapped on a building layout.
    b. Provide tabularized data with individual data and averaged values.

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